A

Mini Project Report

on

EV Charging Station Locator & Finder

Submitted in partial fulfillment of the requirements for the

degree

Second Year Engineering – Computer Science & Engineering (Data Science)

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ABSTRACT

This report presents the development and evaluation of **EV Charging Station Locator & Finder**, an innovative mobile and web-based platform designed to provide real-time access to electric vehicle (EV) charging infrastructure. The system utilizes a combination of GPS-based location tracking, cloud-based backend services, and real-time database synchronization to deliver accurate and up-to-date information about nearby charging stations. The application integrates modern technologies such as Firebase Authentication, Cloud Firestore, and OpenStreetMap APIs to ensure secure user management, efficient data handling, and seamless map-based navigation.

The system architecture includes data acquisition from external APIs, real-time data processing, and dynamic user interaction through an intuitive interface. Advanced functionalities such as live slot availability tracking, booking management, integrated payment systems, and automated slot release mechanisms contribute to improved system efficiency and user convenience. Additional features including user dashboards, booking history tracking, notification systems, and admin monitoring panels enhance usability and system control.

The performance of the system is evaluated using metrics such as response time, real-time data accuracy, system reliability, and user engagement analytics obtained from Firebase. The results demonstrate that the application effectively reduces user effort, minimizes waiting time, and improves decision-making during EV charging. Overall, the system provides a scalable and user-centric solution that supports the growing EV ecosystem and promotes sustainable and efficient transportation.

Keywords--*EV Charging Station Locator, Electric Vehicles (EV), Real-Time Availability, GPS Tracking, Firebase, Cloud Firestore, Flutter, OpenStreetMap API, Slot Booking System, Mobile Application, Payment Integration, User Authentication, Smart Mobility, Sustainable Transportation, Real-Time Data Processing, System Scalability, User Experience, Admin Dashboard, Data Synchronization, EV Infrastructure*

Chapter 1 : Introduction

Introduction

Electric vehicles are rapidly gaining popularity in India as more people are shifting towards eco-friendly and cost-effective modes of transportation. Rising fuel prices and increasing environmental concerns have encouraged users to adopt electric cars and bikes as a better alternative to traditional vehicles. However, despite this growth, one major challenge still exists—the lack of a proper and reliable system to locate nearby charging stations and check their availability in real time. This creates inconvenience for users, especially when they are traveling or in urgent need of charging.

In many cases, EV users face difficulties in finding charging stations and often depend on scattered sources like maps or websites, which may not provide accurate or updated information. As a result, users sometimes travel to a station only to find that all charging slots are occupied, leading to wasted time, effort, and energy. This issue highlights the need for a centralized and efficient solution that can provide complete and real-time information in one place.

The EV Charging Station Locator is here to make things easier. This app uses GPS to help you find charging stations near you. It gives you all the important details you need, like where the station is, if there are any free slots, what type of charger they have, and how much it costs. Everything is in one place, so you can easily find what you're looking for. The app is updated in real-time and is easy to use, making it simpler for people with electric vehicles to charge their cars. With this app, you can just open it up, find a charging station, and get the information you need - it's that simple.

This project is really important for reducing worries about running out of battery power before getting to a charging station. Now, people can plan their trips with confidence, knowing exactly where and when they can charge their cars. The goal is to make owning an electric vehicle easier and more convenient, to use charging stations more efficiently, and to encourage more people to switch to electric vehicles. By doing this, we can help make electric vehicles a more practical choice for everyone.

In real-life situations, users often depend on incomplete or outdated information from

different sources, which does not guarantee accuracy. This leads to situations where users travel to a charging station only to find that it is fully occupied or not functioning properly. Such experiences not only waste time and energy but also reduce confidence in using electric vehicles for long-distance or regular travel. These challenges highlight the need for a unified system that can provide accurate, real-time, and complete information in one place.

The EV Charging Station Locator project is a clever solution to this issue. It's an app that helps people find charging stations near them using GPS. The app gives users a lot of useful information, like whether the stations are available, what types of chargers they have, how much it costs, and more. By putting all this information in one place, the app makes it easier for users to decide where to go. They don't have to look at multiple sources, which can be confusing and time-consuming. This app is a big help to people who need to charge their electric vehicles. It's easy to use and has all the information they need, making it a great tool for anyone with an electric car. With this app, users can quickly find a charging station that meets their needs, saving them time and hassle. The app is a great example of how technology can make our lives easier and more convenient.

1.1 Purpose:

The main purpose of making this system is to help EV users find charging stations without wasting time. Right now there is no single app that shows everything together like station location, slot availability, charger type and price. Users have to search on Google and still they don't get proper real time information.

This app solves that problem. Everything is available in one place. The user just opens the app and within seconds they know which station is free and how to reach there.

The system is also meant to make using electric vehicles more efficient and convenient by cutting down on unnecessary travel and traffic jams that happen when people are looking for charging stations. By directing users to empty slots, it makes better use of the charging infrastructure that is already in place, which reduces the amount of resources that are not being used.

The app also helps electric mobility grow by making the charging process more reliable and easy to use, which encourages more people to switch to electric vehicles. It helps the environment by saving time and energy, and it also helps lower carbon emissions.

The platform also makes prices and services clear, which helps users make smart choices. Beyond these objectives, the system also aims to enhance overall reliability and trust in EV charging services by providing a structured and organized platform instead of scattered information sources.

It focuses on improving user planning by allowing advance booking of charging slots, which helps users manage their journeys more effectively and avoid last-minute issues. The application is also intended to support service providers by offering a digital platform to manage their stations, monitor usage, and improve service quality.

This can lead to better maintenance, reduced downtime, and improved customer satisfaction. Additionally, the system helps in reducing operational inefficiencies by automating processes such as booking, payments, and updates, which minimizes manual intervention.

Another important purpose is to create a connected ecosystem where users and charging stations interact seamlessly, improving coordination between demand and availability. Beyond these aspects, the system also aims to improve **accessibility and inclusiveness** by making EV charging information easily available to all types of users, including first-time EV owners who may not be familiar with charging networks. It reduces dependency on technical knowledge by providing a simple and guided interface that anyone can understand and use.

The application is also intended to enhance **safety and reliability** by helping users avoid situations where their vehicle battery becomes critically low without access to a charging station. By providing timely information and navigation support, it ensures a more secure travel experience. Another important purpose is to enable **better time management** for users. Instead of waiting in long queues or searching for available stations, users can plan their charging schedules efficiently and utilize their time more productively

1.2 Problem Statement:

India's electric vehicle (EV) ecosystem is growing rapidly, but it still faces several critical challenges that limit its full potential. One of the major problems is the lack of a unified and reliable system to locate EV charging stations with accurate, real-time information. Most users currently depend on scattered sources like maps, websites, or individual service provider apps, which often provide incomplete or outdated data. This makes it difficult for users to determine key details such as station availability, charger type compatibility, pricing, and operational status.

Traditional approaches are inefficient, as users may have to manually search multiple platforms or physically visit stations to confirm availability. For example, a charging station may appear nearby on a map but could be fully occupied, under maintenance, or incompatible with a user's vehicle—factors that are not always reflected in basic systems. This results in wasted time, increased traffic congestion, and a frustrating user experience. Additionally, the absence of real-time updates and predictive insights makes it harder for users to plan their trips effectively.

Another major issue is the lack of integration across different services. Users must switch between apps or platforms for navigation, availability, booking, and payment, leading to a fragmented experience. There are also concerns related to data security and personalization, as many existing solutions do not provide secure authentication or user-specific features like booking history and preferences. Furthermore, the absence of proper visualization tools and user-friendly interfaces makes it difficult for non-technical users to understand and utilize the available information effectively.

From an infrastructure perspective, there are challenges related to scalability and system efficiency. Many existing solutions are not designed to handle large-scale real-time data or integrate seamlessly with multiple charging networks. This leads to poor coordination between demand and availability, resulting in underutilized stations in some areas and overcrowding in others. As a result, despite increasing EV adoption, the charging infrastructure is not used optimally.

The EV Charging Station Locator project addresses these challenges by providing an

integrated, real-time, and user-friendly platform. It consolidates all essential information—such as location, slot availability, charger types, and pricing—into a single system, eliminating the need to rely on multiple sources. The application uses modern technologies to ensure real-time updates, secure authentication, and efficient data handling, providing users with accurate and reliable information at all times.

The system is designed to be easy to use, with a simple and intuitive interface that makes it accessible to everyone, regardless of their technical background. This means that both tech-savvy and non-tech-savvy users can navigate the system with ease. Additionally, the system is built to be scalable, allowing it to grow and expand as needed, and can be integrated with multiple charging providers. It also includes advanced features like predictive availability and smart recommendations, which help to make the user experience even more seamless. By combining real-time data, smooth functionality, and a user-centric design, the project aims to break down the current barriers to widespread adoption of electric vehicles, including technical, usability, and accessibility issues. The ultimate goal is to create a more efficient and sustainable transportation system for the future. With its focus on usability and scalability, the system has the potential to make a significant impact on the way we travel, and could play a key role in shaping the future of transportation.

1.3 Objectives

This project is designed to address the practical challenges faced by electric vehicle (EV) users in locating and accessing charging stations efficiently. It focuses on improving convenience, reducing uncertainty, and enhancing the overall charging experience through real-time data, seamless booking, and integrated services. By combining GPS technology, real-time database updates, and user-friendly design, the system provides a complete solution that supports users in managing their charging needs with ease and confidence.

1. To enable GPS-based nearby charging station detection

The primary objective is to allow EV users to easily find nearby charging stations using GPS technology. The system automatically detects the user's current location and displays a list or map view of available stations in close proximity. This eliminates the need for manual searching and ensures quick access to relevant information. By providing accurate location-based results, the application helps users save time and navigate efficiently, especially in unfamiliar areas.

2. To provide real-time slot availability information

This objective focuses on delivering up-to-date information about charging slot availability at each station. The system continuously updates data from the backend to reflect the current status of slots, whether they are available, occupied, or reserved. This real-time functionality ensures that users can make informed decisions before traveling to a station, thereby reducing waiting time and avoiding unnecessary trips.

3. To implement advance slot booking functionality

The application enables users to book charging slots in advance according to their preferred time and location. This feature helps users plan their journeys more effectively and guarantees availability upon arrival. By reducing uncertainty and waiting queues,

advance booking improves convenience and ensures a smoother charging experience, particularly during peak usage hours.

4. To integrate secure in-app payment system

Another key objective is to provide a seamless and secure payment mechanism within the application. Users can complete transactions directly through the app without needing external platforms or physical payment methods. This not only simplifies the process but also enhances transparency, allowing users to view pricing details and payment history easily.

5. To enhance overall user experience and reduce stress

The final objective is to create a simple, efficient, and user-friendly system that minimizes the challenges associated with EV charging. By integrating all essential features—location tracking, real-time updates, booking, and payment—into one platform, the application reduces confusion and effort for users. The goal is to make EV usage more convenient, reliable, and stress-free, encouraging wider adoption of electric vehicles.

1.4 Scope

The application was designed to function as a mobile phone application for Android, as well as for web applications. In this case, the main functions include location tracking, finding stations near your location, finding empty slots, booking the slot, and paying for the slot. More sophisticated features, such as route tracking and prediction of which station would become empty after some time, were not included in the current application. They could have been implemented later.

Currently, the application offers basic features including location tracking, nearby station finding, slot availability check, booking, and payments. But the scope is wide open and vast in terms of improving efficiency and user experience, as well as enhancing scalability.

In the future, more intelligent features such as real-time routing analysis could be used to guide users towards the most convenient and less crowded station. The application will utilize machine learning based predictions of the slot availability based on previous usages and inform users ahead of time about the same.

Dynamic pricing based on demand could also be used for slots, meaning slot costs will differ depending on time, location, and demand, much like the ridesharing applications. The AI engine will recommend users the best slot, taking into account distance, cost, and availability.

Integration with IoT enabled smart city systems will also be used to communicate with parking/charging systems and other smart devices within the cities. Automatic detection of slot availability through the sensor system and real-time occupancy monitoring are some of the improvements that could be made in the future.

Chapter 2: Proposed System

2.1 Features and Functionality

- **GPS Location Tracking**

This feature uses GPS to figure out where you are and then gives you information that's relevant to your location. It makes sure that everything you see is tailored to where you are right now, so you get the most accurate and helpful results.

- **Nearby Charging Station Search**

The app helps you find charging stations near you by showing them on a list or map. This way, you can easily see which ones are closest and available, without having to search around. It uses your location to do this, making it quick and simple to find a place to charge your electric vehicle.

- **Real-Time Slot Availability**

The system keeps track of the charging slots and lets users know if they are free, being used, or already booked. This information helps people plan ahead before heading to a charging station.

- **Charging Type Display**

The app gives you information about the different kinds of chargers you can find at each station, like fast charging or slow charging. This helps you pick a station that works with your car.

- **Navigation / Map Integration**

This feature is really helpful because it uses maps to give you directions from where you are to the charging station you've chosen. It makes it easy to get there without getting lost.

- **Slot Booking Option**

You can book a charging spot ahead of time using the app, which cuts down on wait time and makes sure a spot is open when you get there. This way, you don't have to worry about showing up and finding out that all the charging spots are taken.

- **Online Payment System**

The app lets you pay for charging right inside it, making things fast, safe, and easy - no need to use other ways to pay.

- **User Registration & Login**

This feature lets users make their own accounts and log in safely. It's useful for keeping track of personal info, what you've booked in the past, and what you like.

- **Ratings and Reviews**

Users can rate and review charging stations based on their experience. This helps other users make better decisions and improves service quality.

- **Notifications**

The app keeps you in the loop by sending you alerts and updates. You'll get messages about things like booking confirmations, how your payments are going, reminders about your time slots, and any important announcements. This way, you're always informed about what's going on.

The system also helps the environment by reducing energy waste and lowering carbon emissions through efficient route planning and less idle time. It improves transparency by clearly showing important details like charger type, pricing, availability, and estimated waiting time. This allows users to make informed and cost-effective choices.

Additionally, the platform can assist service providers and government authorities by gathering and examining real-time and historical data. This data can help with better infrastructure planning, predicting demand, and shaping policies. Overall, the system aims to play a vital role in developing smart cities by combining digital technology with modern transportation needs. It seeks to create a more connected, efficient, and sustainable mobility ecosystem.

Chapter 3 : Project Outcomes

1. User can find nearest charging station within few seconds:

The application offers a simple and user-friendly solution for electric vehicle users by making the charging process easier. Users can quickly locate the nearest charging station in just a few seconds. This removes the need for manual searches or relying on incomplete information. There's no need to call or visit stations to check availability. The system provides real-time data about open slots, charger types, and pricing. This saves time and effort while helping users make better decisions.

Additionally, the application allows users to complete the whole process—from selecting a station to booking a slot and making payments—on one platform. This ensures a smooth experience. Real-time updates mean that users always have correct and up-to-date information, which cuts down on uncertainty and waiting time. By streamlining these processes, the system reduces the stress and confusion that often come with EV charging, especially for new users.

2. No need to guess or call the station to check availability:

Moreover, the application improves convenience by reducing unnecessary travel and optimizing route selection. This leads to better time management and energy efficiency. It also enhances the use of charging stations by directing users to available slots, preventing overcrowding. Overall, the system makes electric vehicle use more reliable, accessible, and comfortable, making it easier for more people to adopt EVs.

3. Booking and payment both happen in the app without going anywhere else:

The application simplifies finding and booking charging stations. It also significantly improves the experience of using electric vehicles. With instant access to nearby stations, users save time that they would otherwise spend searching or waiting. By eliminating the need to call stations to check availability, the system becomes more reliable and efficient. Integrated booking and payment features allow users to plan their charging sessions in advance, avoiding last-minute issues and ensuring a smooth journey.

4. Real time data means user always gets correct information:

Additionally, having real-time data enhances accuracy and trust in the system. Users can rely on the displayed information without second-guessing. This reduces anxiety, especially when battery levels are low and quick decisions are necessary. The application also aids in better time management by minimizing waiting queues and directing users to available slots. Frequent users benefit from a consistent and predictable charging routine.

5. EV usage becomes much less stressful:

Furthermore, the system supports environmental sustainability by cutting down on unnecessary vehicle movement and energy waste. It promotes fair and efficient use of charging stations by spreading demand across different locations.

Chapter 4: Software Requirements:

- **Frontend:** Flutter SDK
- **Backend:** Firebase
- **Database:** Cloud Firestore
- **Tools:** Android studio/VS code
- **Language**: Dart
- **Platform:** Android/ web

Frontend (Flutter SDK): Used to build the user interface of the application. Flutter SDK allows development of fast, responsive, and visually appealing apps for both mobile and web using a single codebase.

Backend (Firebase): Firebase is used to handle server-side operations like authentication, real-time data syncing, and hosting, making the app scalable and efficient without managing servers.

Database (Cloud Firestore): Cloud Firestore stores user data, station details, bookings, and availability in a real-time NoSQL database, ensuring fast and secure data access.

Tools (Android Studio / VS Code): Android Studio and Visual Studio Code are used for coding, debugging, and testing the application efficiently.

Language (Dart): Dart is the core programming language used to develop the app, known for its speed and smooth integration with Flutter.

Platform (Android / Web): The application is designed to run on both mobile devices and web browsers, ensuring accessibility across different devices and increasing user reach.

The application is developed using the Flutter SDK, which enables building a fast and responsive user interface for both mobile and web platforms using a single codebase. The backend is powered by Firebase, which manages authentication, real-time data synchronization, and server-side operations efficiently without the need for dedicated servers. Data is stored in Cloud Firestore, a real-time NoSQL database that ensures quick and secure access to information such as user details, charging stations, and bookings. The application is developed using the Dart programming language and is built and tested using tools like Android Studio and Visual Studio Code. Overall, the system is designed to run on both Android devices and web browsers, making it widely accessible and easy to use.

CHAPTER 5: Project Design

5.1 Use Case diagram:

Figure 5.1 shows the Use Case Diagram for the EV Charge Finder application, outlining interactions between the User and Admin actors with the system. It includes key functions such as login and authentication, exploring nearby EV charging stations on an interactive map, searching stations across cities, viewing real-time slot availability, and booking a charging slot by selecting date, time, and duration. The diagram also highlights system extensions such as payment processing via UPI, card, or cash, automatic slot release after session expiry, booking confirmation generation, and admin dashboard monitoring with booking statistics, revenue tracking, and real-time slot management using Firebase and OpenStreetMap.

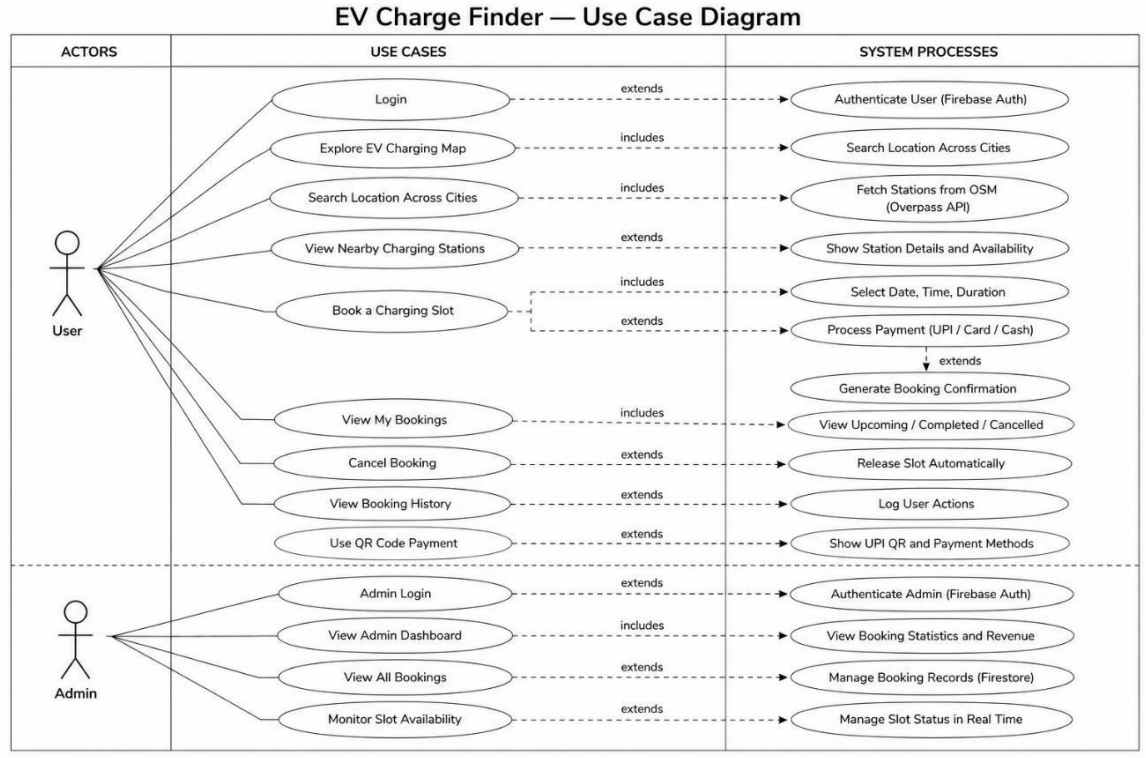

Fig 5.1: Use Case Diagram

Actors:

- User
- Admin

Use Cases and connections:

USER side:

- Login → extends → Authenticate User (Firebase Auth)
- Explore EV Charging Map → includes → Search Location Across Cities
- Search Location Across Cities → Fetch Stations from OSM (Overpass API)
- View Nearby Charging Stations → extends → Show Station Details and Availability
- Book a Charging Slot → includes → Select Date, Time, Duration
- Book a Charging Slot → extends → Process Payment (UPI / Card / Cash)
- Process Payment (UPI / Card / Cash) → extends → Generate Booking Confirmation
- View My Bookings → includes → View Upcoming / Completed / Cancelled
- Cancel Booking → extends → Release Slot Automatically
- View Booking History → extends → Log User Actions
- Use QR Code Payment → extends → Show UPI QR and Payment Methods

ADMIN side:

- Admin Login → extends → Authenticate Admin (Firebase Auth)
- View Admin Dashboard → includes → View Booking Statistics and Revenue
- View All Bookings → extends → Manage Booking Records (Firestore)
- Monitor Slot Availability → extends → Manage Slot Status in Real Time

5.2 DFD (Data Flow Diagram):

Figure 5.2 illustrates the Data Flow Diagram of the EV Charge Finder application, outlining the sequential flow of data from user interaction to system response. The process begins with the user opening the app and granting location permission, followed by the system fetching nearby EV charging stations from the OpenStreetMap Overpass API based on the user's current coordinates. The fetched station data is processed, parsed, and stored in the Firebase Firestore database. The user can then search for stations in other cities, select a station, choose a date, time slot, and charging duration, and proceed to payment via UPI, card, or cash. Upon successful payment, the booking record is saved in Firestore and a confirmation is displayed to the user. The system automatically releases the booked slot after the session duration expires and updates the booking status to completed, which is reflected in the user's booking history and the admin dashboard in real time.

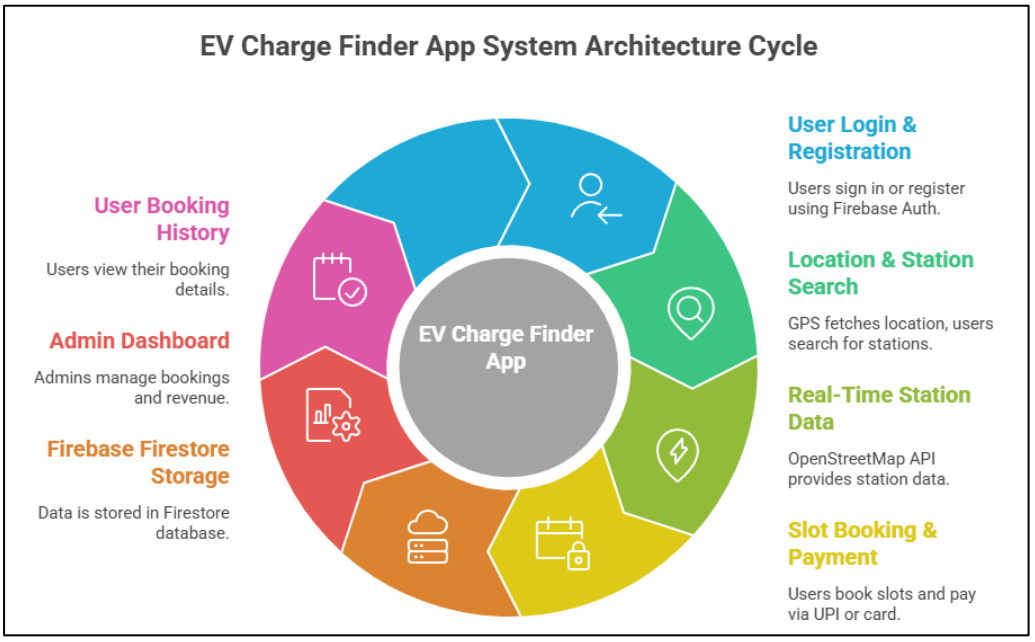

Fig 5.2: Data Flow Diagram

5.3 System Architecture:

Figure 3.3 illustrates the overall system architecture of the EV Charge Finder application, showing how data flows through the system from user input to final output. The workflow starts with the user providing input such as GPS location, city search query, login credentials, and slot selection details. This input is passed to the data processing layer, where Firebase Authentication handles secure login, the OpenStreetMap Overpass API fetches and parses nearby EV charging station data, and Firebase Firestore manages all booking records and real-time slot status updates. The processed data then produces outputs including the interactive station map, real-time slot availability, booking confirmation, and the admin dashboard with revenue and booking statistics. The architecture also includes an auto-release feedback loop, where the system automatically updates the slot status to available and marks the booking as completed once the selected charging session duration expires, ensuring accurate real-time data at all times.

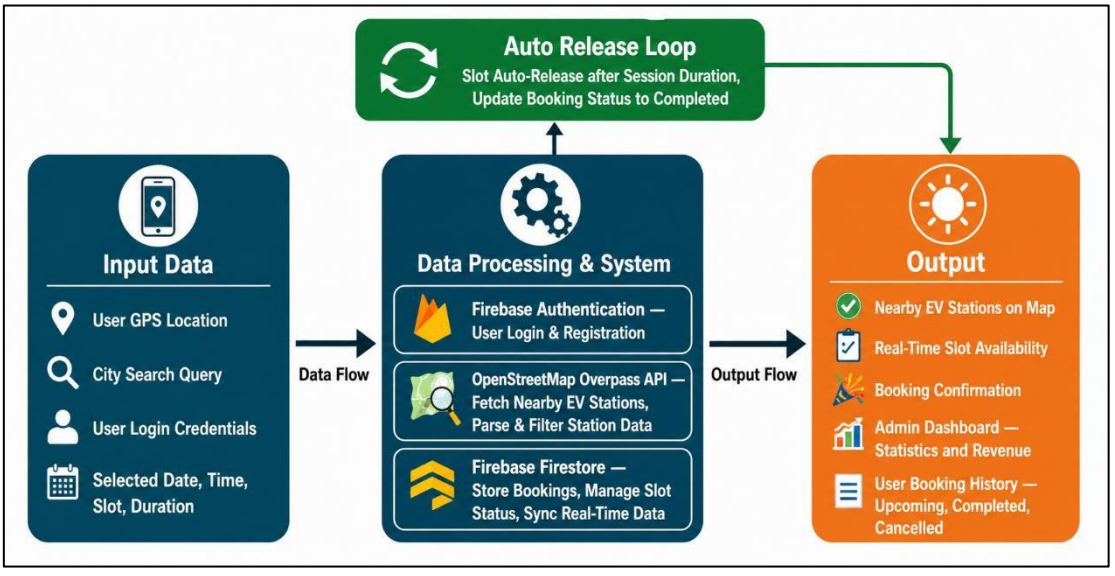

Fig 5.3: System Architecture

1. User Input Collection:

- Input includes the user's GPS location, city search query, login credentials, and selected date, time slot, and charging duration.

2. Authentication and Login:

- Firebase Authentication verifies user credentials securely.
- New users can register while existing users sign in directly to access all app

features.

3. Station Data Fetching:

- The OpenStreetMap Overpass API fetches nearby EV charging stations within 50km based on the user's location.

4. Booking and Payment:

- The user selects a station, date, time slot, and duration and proceeds to payment via UPI, card, or cash.

5. Auto-Release Feedback Loop:

- Once the session duration expires, the system automatically releases the booked slot.
- Booking status is updated to completed and the slot becomes available for other users in real time.

6. Output Generation:

- The system outputs:
  - Nearby EV stations on an interactive map
  - Real-time slot availability per station
  - Booking confirmation with complete details
  - Admin dashboard with bookings, revenue, and slot status
  - User booking history with upcoming, completed, and cancelled bookings

5.4 Implementation:

The implementation of the EV Charge Finder application focused on developing a robust, secure, and intuitive mobile application to help users locate nearby EV charging stations and book charging slots in real time. Key aspects of the implementation included integrating GPS-based location detection, fetching live station data from OpenStreetMap, ensuring smooth slot booking and payment flow, and implementing Firebase Authentication for secure user access. The application was built using Flutter for a clean and responsive cross-platform mobile experience, with Firebase Firestore handling all real-time data storage and synchronization.

Figure 5.4.1: Login Screen

Figure 5.4.1 shows the Login Screen of the EV Charge Finder application, which serves as the secure entry point for all users. The screen provides three options — Sign In, Register, and Admin — allowing both regular users and administrators to access the application. Firebase Authentication is used in the backend to verify user credentials and ensure secure access to all features of the application.

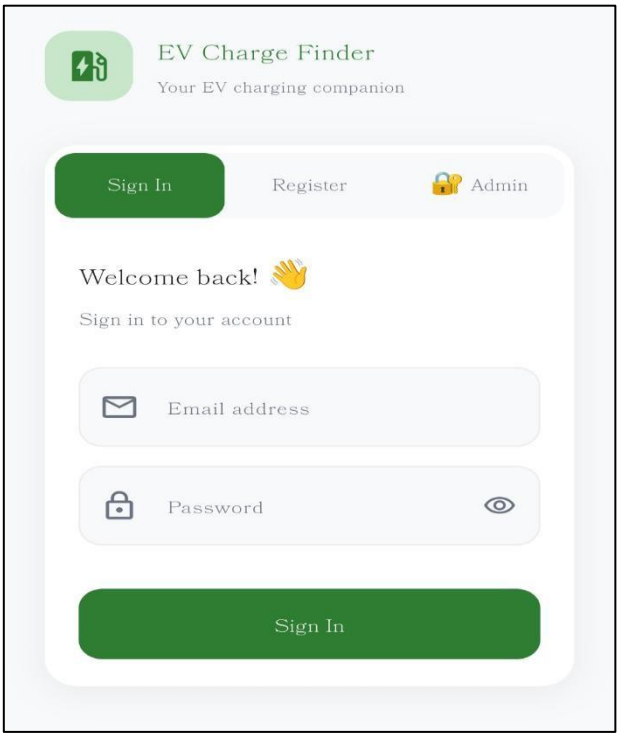

Figure 5.4.1: Login Screen

Figure 5.4.2: Home Map Screen

Figure 5.4.2 shows the Home Map Screen displayed immediately after a successful login. The screen renders an interactive OpenStreetMap with the user's current GPS location marked with a blue dot at the center. The map automatically detects the user's real-time location and is ready to fetch and display nearby EV charging stations, serving as the main navigation hub of the application.

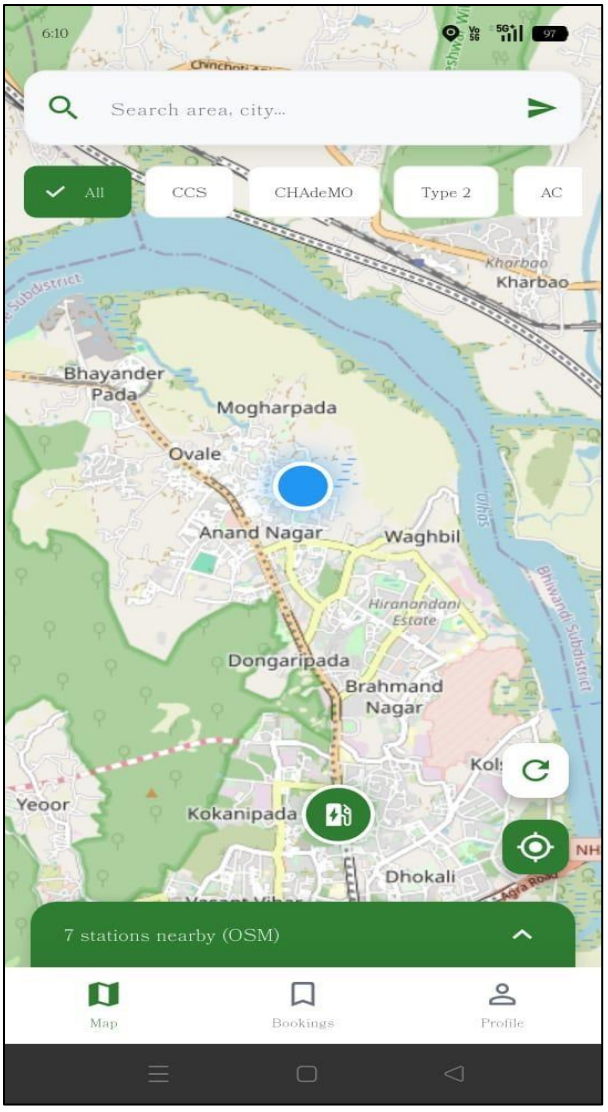

Figure 5.4.2: Home Map Screen

Figure 5.4.3: Nearby Stations List

Figure 5.4.3 shows the Nearby Stations Screen displaying the list of EV charging stations fetched in real time from the OpenStreetMap Overpass API. A total of 7 stations are found nearby, with each station showing its name, distance, number of charging points, charger type, and available free slots. Stations like Fortum EV Charger, Tata Power, and Delta are listed with their distances and availability, confirming that real-time station fetching is working correctly.

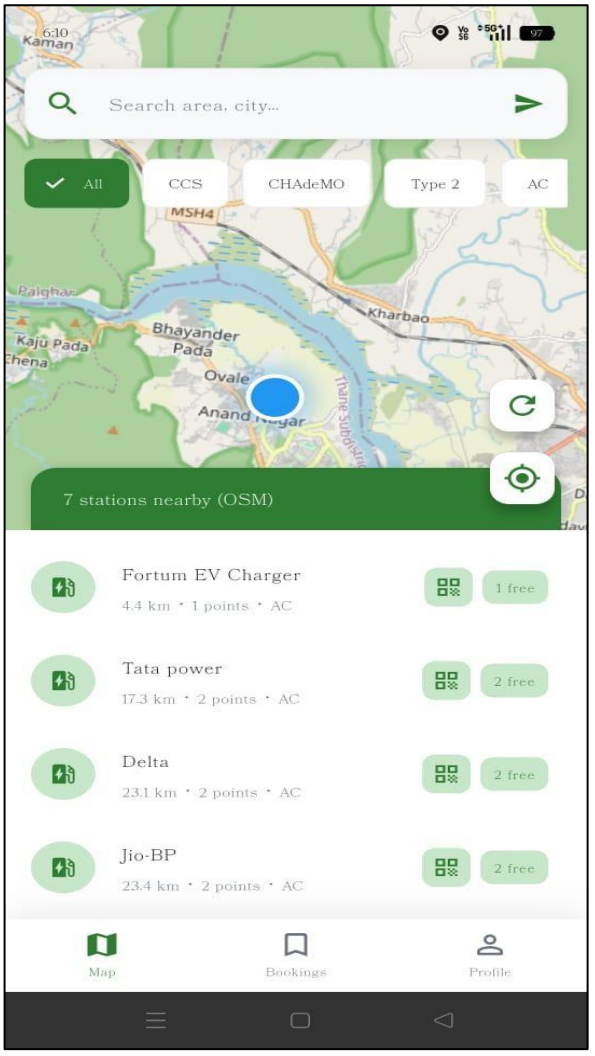

Figure 5.4.3: Nearby Stations List

Figure 5.4.4: Slot and Duration Selection Screen

Figure 5.4.4 shows the Booking Screen where the user selects the charging duration, date, and time slot. Four duration options are available — 1 min, 30 min, 1 hour, and 2 hours — with the estimated cost displayed below. The date picker allows selecting any date within 30 days, and the time slot grid shows available slots in green and booked slots in red, allowing the user to clearly identify and select a free slot for their booking.

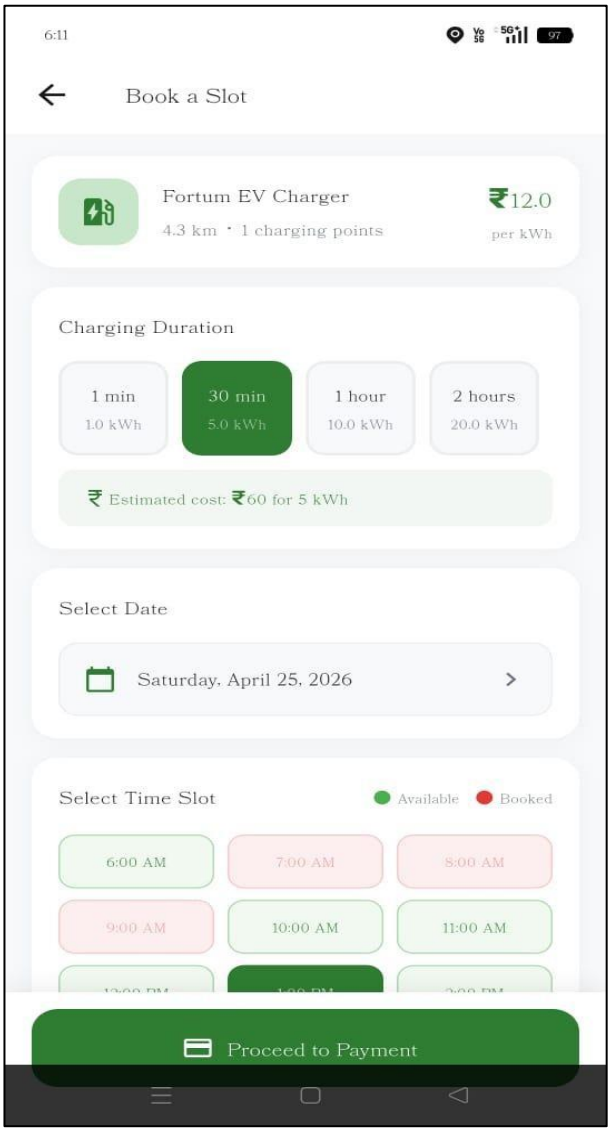

Figure 5.4.4: Slot and Duration Selection Screen

Figure 5.4.5: Payment Screen

Figure 5.4.5 shows the Payment Screen displaying the complete order summary and payment method selection. The summary confirms the station, date, time slot, duration, rate, and total amount of ₹60. Three payment options are available — UPI, Credit/Debit Card, and Net Banking — with UPI selected and highlighted, confirming that the complete payment flow is correctly implemented.

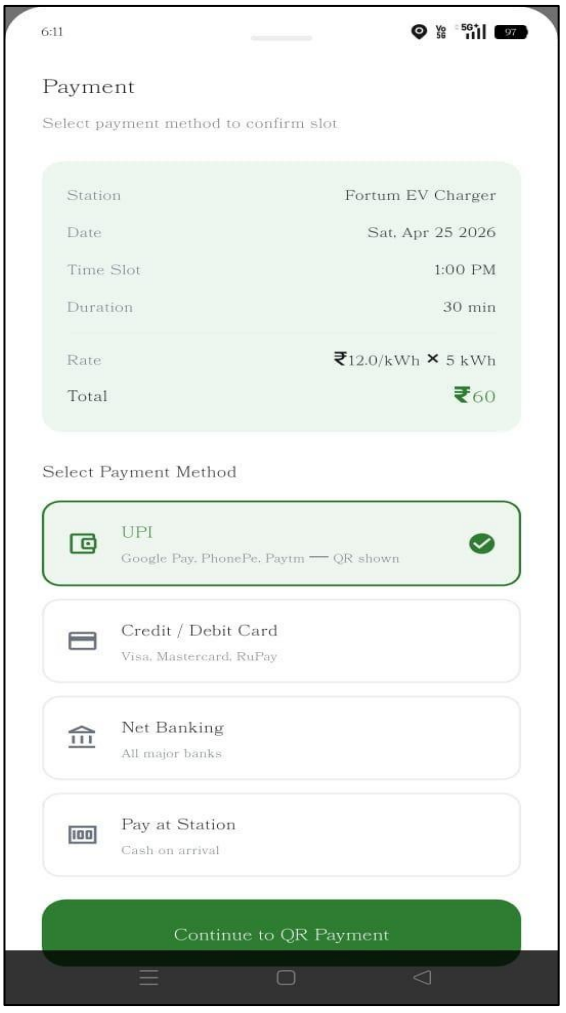

Figure 5.4.5: Payment Screen

Figure 5.4.6: UPI QR Payment Screen

Figure 5.4.6 shows the UPI QR Payment Screen where users can scan and pay using any UPI application. The screen displays the total amount of ₹60, the station name Fortum EV Charger, a scannable QR code, and the UPI ID evcharge@ybl. Payment options including Google Pay, PhonePe, and Paytm are supported, confirming that the QR-based payment feature is fully functional.

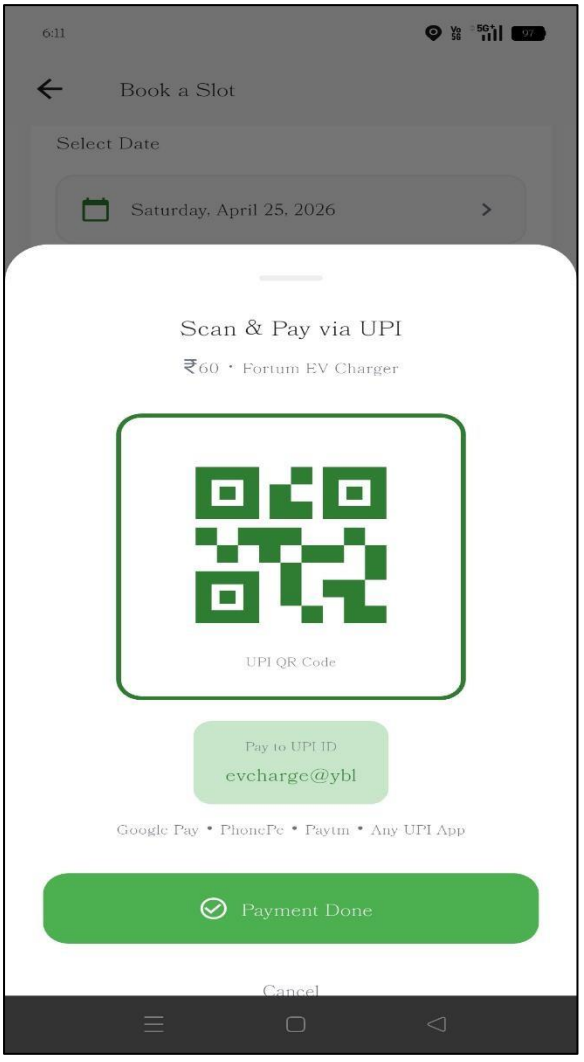

Figure 5.4.6: UPI QR Payment Screen

Figure 5.4.7: Booking Confirmation Screen

Figure 5.4.7 shows the Booking Confirmation Screen displayed after a successful payment. The screen shows a green tick with the message "Booking Confirmed" along with complete booking details including the Booking ID, station name as Fortum EV Charger, date as Saturday April 25, time as 1:00 PM, duration as 30 minutes, and amount paid as ₹60. The screen also displays a note that the slot will auto-release after 30 minutes, confirming the complete booking flow is working end to end.

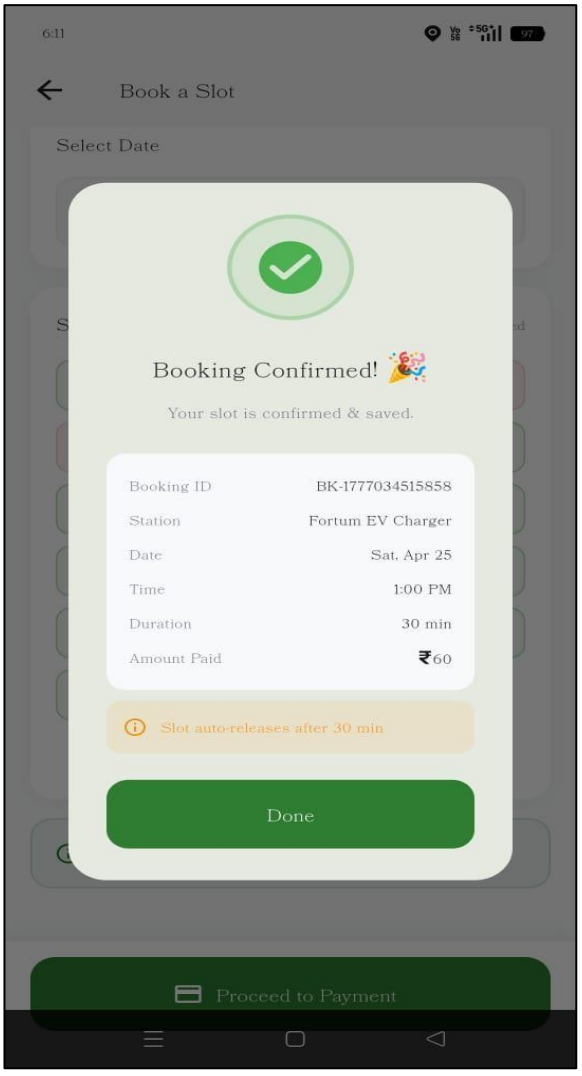

Figure 5.4.7: Booking Confirmation Screen

Figure 5.4.8: My Bookings Screen — Upcoming and Completed

Figure 5.4.8 shows the My Bookings Screen displaying both upcoming and completed bookings for the logged-in user. The upcoming tab shows active bookings at Tata Power and Fortum EV Charger with their respective dates, times, durations, and amounts paid. The completed tab shows bookings at Ather Energy and Delta that have been automatically marked as completed after the session duration expired, confirming that the auto-release and status update feature is working correctly in real time.

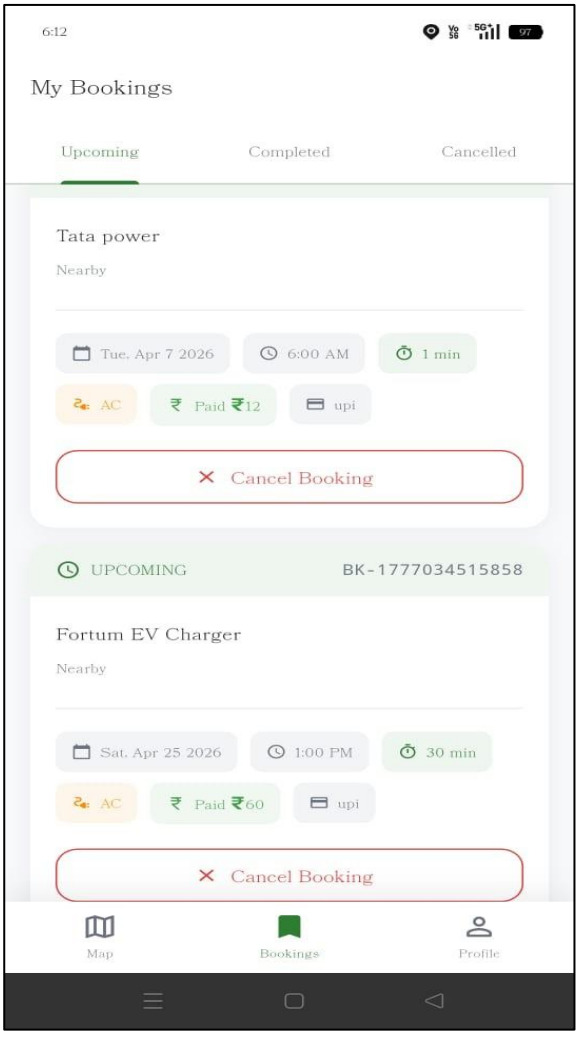

Figure 5.4.8: My Bookings Screen

Figure 5.4.9: Booking History Screen

Figure 5.4.9 shows the Booking History Screen displaying all past bookings made by the logged-in user. The screen shows bookings at Fortum EV Charger and Tata Power with complete details including date, time, duration, amount paid, and unique booking ID. This screen confirms that all booking records are correctly saved in Firebase Firestore and accurately fetched and displayed for the specific logged-in user, ensuring data privacy and correctness.

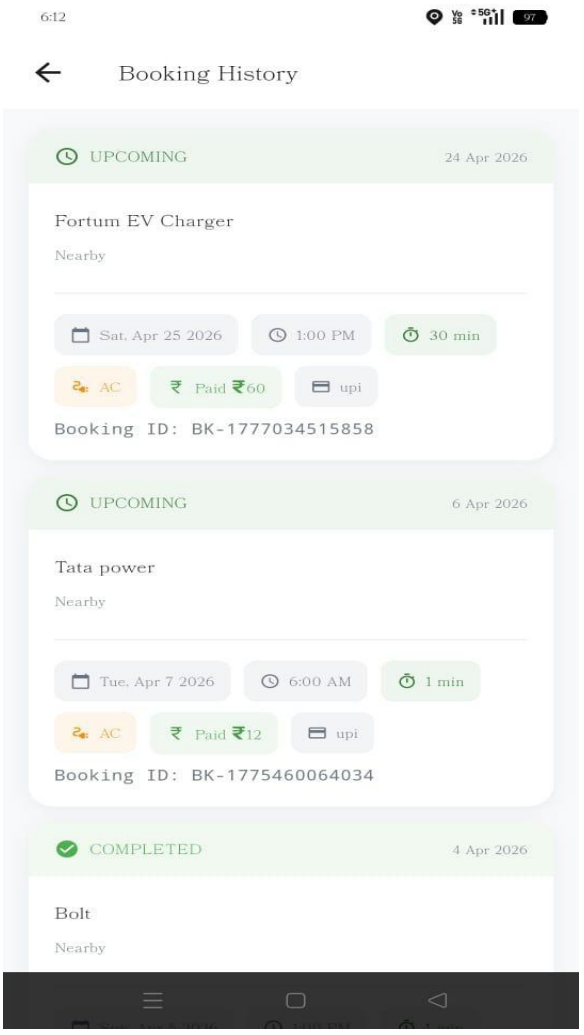

Figure 5.4.9: Booking History Screen

Figure 5.4.10 & 11: Admin Login and Admin Dashboard Screen

Figure 5.4.10 & 11 shows the Admin Login Screen and Admin Dashboard of the EV Charge Finder application. The Admin Login Screen is accessible by selecting the Admin tab on the main login page, where station owners and administrators must enter a unique admin access code to gain access to the admin panel. This separate authentication layer ensures that regular users cannot access administrative features, keeping all booking records, revenue data, and slot management restricted to authorized personnel only. Once logged in successfully, the Admin Dashboard displays all bookings made across all users of the application, with each entry showing the station name, charger type, amount paid, time slot, unique booking ID, and current status marked as either Upcoming or Completed. Bookings at Fortum EV Charger and Tata Power are shown as Upcoming while bookings at Bok and Delta are marked as Completed, confirming that the admin monitoring feature is fully functional and providing complete real-time visibility over all bookings and revenue generated through the application.

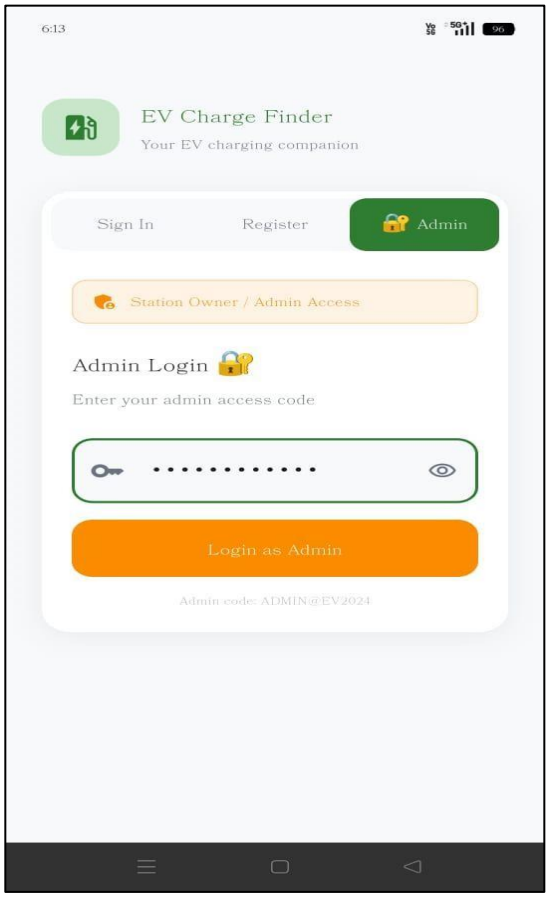

Figure 5.4.10: Admin Login Screen

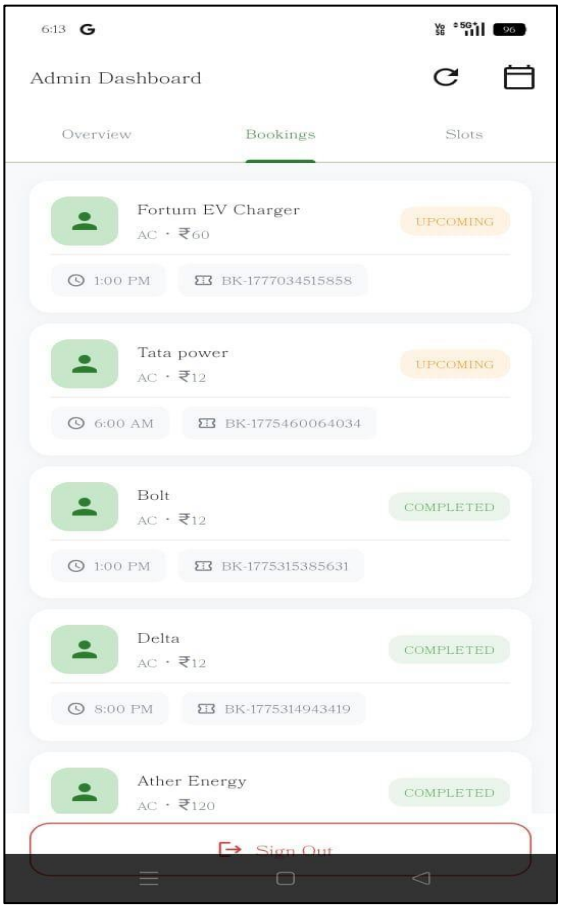

Figure 5.4.11: Admin Dashboard

Chapter 6: Technical Specification

The EV Charge Finder application is built using a modern mobile technology stack that ensures real-time performance, secure user management, and seamless data handling across all features of the application.

Frontend — Flutter (Dart)

The user interface is built using the Flutter SDK, which provides a clean, responsive, and cross-platform mobile application that runs smoothly on both Android and iOS devices. Flutter enables fast UI rendering and a consistent user experience across all screen sizes.

Backend — Firebase Services

Firebase acts as the complete backend for the application, handling all server-side operations without the need for a separate server setup. It manages real-time data synchronization, user authentication, and cloud database operations efficiently.

Database — Firebase Firestore

Cloud Firestore is used as the primary database to store all application data including user profiles, booking records, charging station details, and real-time slot availability. Firestore enables instant data updates across all users without manual refresh.

Authentication — Firebase Authentication

Firebase Authentication handles secure user login and registration using email and password. It ensures only authorized users can access booking and payment features, while the admin login is separately managed with a unique access code.

Map and Station Data — OpenStreetMap Overpass API

Real-time EV charging station data is fetched from the OpenStreetMap Overpass API based on the user's GPS location or searched city. The API returns nearby stations within 50km which are then parsed, filtered, and displayed on the interactive map.

Payment — UPI, Card, and Cash

The application supports multiple payment methods including UPI with QR code scanning, credit or debit card, net banking, and cash payment at the station, giving users full flexibility in completing their booking payment.

Chapter 7: Project Scheduling

Gantt Chart Description

The project development is scheduled from January to March covering all phases from requirement analysis to final documentation

Month 1 (Jan 10 – Early Feb) → Planning & Design

Focus: Initial setup and app design

- Project Setup
- UI/UX Design
- Navigation & Structure Type: Design & Development

All foundational work completed

Month 2 (Feb – Early Mar) → Core Features Development

Focus: Building main functionalities

- Location Services (GPS)
- Map Integration
- Nearby Charging Stations
- Search & Filters

Type: Core Functional Features Core app features fully developed

Month 3 (Mid Mar – Early Apr) → Integration & Features Enhancement

Focus: Backend and advanced features

- Firebase Integration
- Real-time Data Fetch
- User Features (Favorites) Type: Integration

Backend successfully connected

Month 4 (Apr) → Testing & Deployment

Focus: Final quality check and release

- Testing & Bug Fixing

- Deployment Type: Testing Type: Deployment

App finalized and launched

EV CHARGER FINDER MONTHLY TASK GANTT CHART

Project Name:	EV Charger Finder	Start Date:	10-Jan
Project Lead:	Gargi Warke	End Date:	09-Apr

ACTIVITY	% DONE	START	END	DURATION	STATUS	Month 1				Month 2				Month 3				Month 4			
						W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13			
PHASE 1																					
Project Setup	100%	10-Jan	17-Jan	1W	Done																
UI/UX Design	100%	17-Jan	27-Jan	2W	Done																
Navigation & Structure	100%	27-Jan	05-Feb	2W	Done																
Location Services	100%	05-Feb	15-Feb	2W	Done																
Map Integration	100%	15-Feb	25-Feb	2W	Done																
Nearby Charging Stations	100%	25-Feb	05-Mar	2W	Done																
Search & Filters	100%	05-Mar	15-Mar	2W	Done																
PHASE 2																					
Firebase Integration	100%	15-Mar	20-Mar	1W	Done																
Real-time Data Fetch	100%	20-Mar	28-Mar	1W	Done																
User Features (Favorites)	100%	28-Mar	02-Apr	1W	Done																
Testing & Bug Fixing	100%	02-Apr	07-Apr	1W	Done																
Deployment	100%	07-Apr	09-Apr	1W	Done																

Legend:

Color	Meaning	Used For
Blue	Design & Development	UI/UX, Navigation, Core Features
Teal	Core Functional Features	GPS, Map, Search
Orange	Integration	Firebase, APIs
Yellow	Testing Phase	Bug fixing, QA
Dark Gray	Deployment	Final release

Fig 7.1: Gantt Chart

Chapter 8: Results

The development and testing of the EV Charge Finder application produced significant outcomes across thefrontend, backend, and real-time data modules. The results demonstrate the application's ability to accurately fetch nearby EV charging stations, manage slot bookings in real time, process payments smoothly, and provide a clean and intuitive user experience. The Firebase Analytics data and Firestore database records further confirm that the application is fully functional, stable, and successfully tested on real Android devices in a real-world environment.

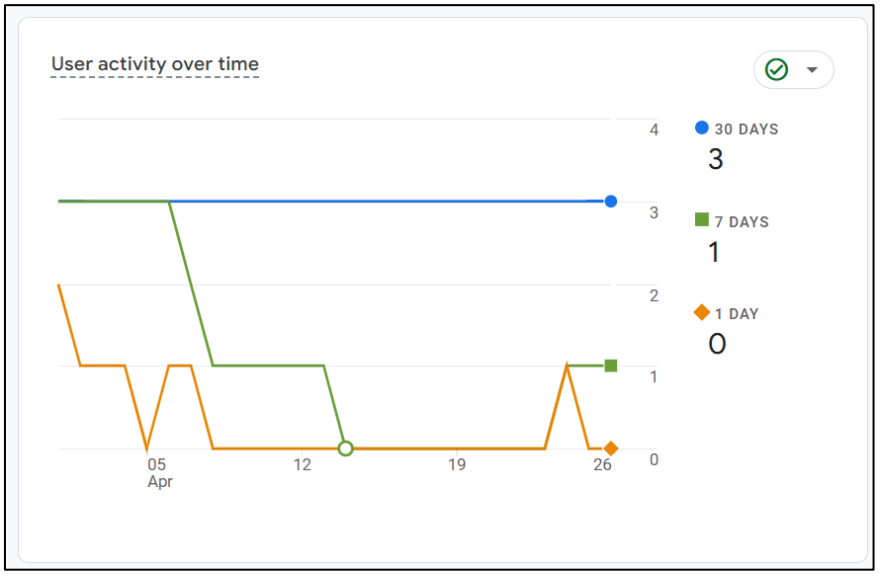

Fig 8.1: User Activity Over Time

Figure 8.1 shows the user activity recorded over the last 30 days on Firebase Analytics for the EV Charge Finder application.

- Total of 3 unique users accessed the application over the 30-day period
- 1 active user recorded in the last 7 days confirming ongoing usage
- Activity pattern visible across April confirms app was running on real devices
- Validates that the application is successfully deployed and functional in a real-world environment

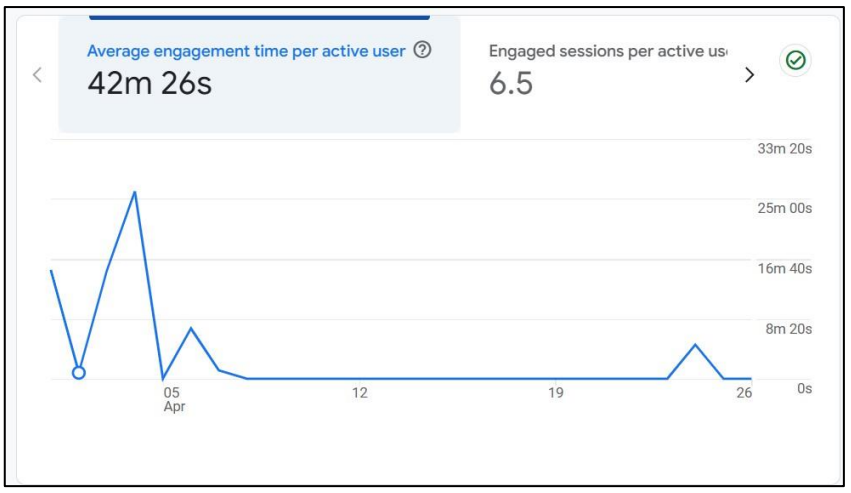

Fig 8.2: Average Engagement Time

Figure 8.2 shows the average engagement time and session statistics recorded by Firebase Analytics.

- Average engagement time of 42 minutes 26 seconds per active user recorded
- 6.5 engaged sessions per active user confirming repeated app usage
- Peak engagement visible in early April during active testing period
- Confirms that all major features including map, booking, payment, and history are working correctly and keeping users engaged

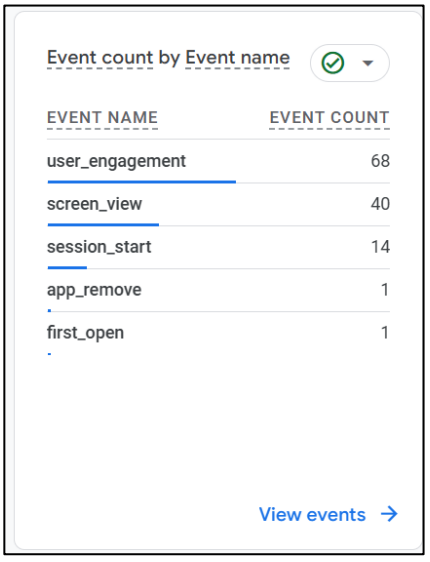

Fig 8.3: Event Count by Event Name

Figure 8.3 shows the event tracking data recorded by Firebase Analytics during the testing period.

- 68 user engagement events recorded confirming active interaction throughout sessions
- 40 screen view events proving users navigated through multiple screens including map, booking, payment, and history
- 14 session start events confirming users opened the application across multiple separate sessions
- 1 first open event confirming successful installation on a real device
- Overall confirms complete end-to-end functionality of the application is working correctly

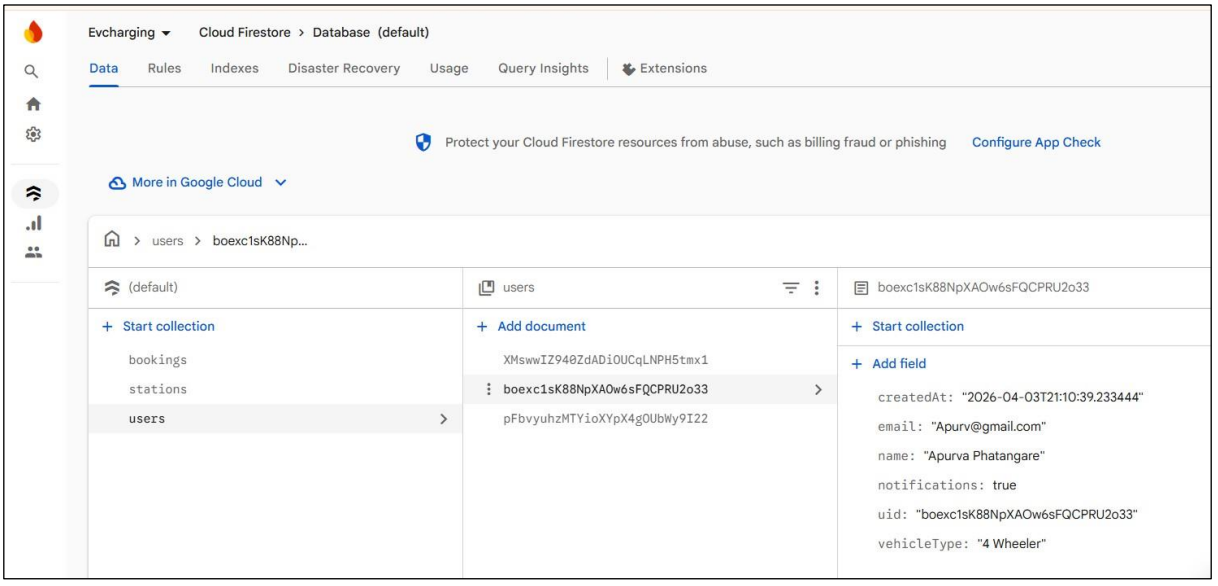

Fig 8.4: Firebase Firestore Database — Users Collection

Figure 8.4 shows the Firebase Cloud Firestore database structure and users collection for the EV Charge Finder application.

- Firestore contains 3 main collections — bookings, stations, and users — confirming complete database structure is correctly set up
- user documents are stored in the users collection confirming successful registrations
- Opened user document confirms all required fields are correctly saved including name, email, uid, vehicleType, notifications, and createdAt timestamp
- Confirms that user registration and Firestore data storage is working perfectly in real time

Chapter 9: Conclusion

The EV Charging Station Locator & Finder system successfully addresses one of the major challenges faced by electric vehicle (EV) users—efficiently locating and accessing reliable charging infrastructure. With the rapid growth of EV adoption, the need for a centralized, accurate, and real-time solution has become increasingly important. This project fulfills that need by integrating advanced technologies such as GPS-based location tracking, real-time database synchronization, and cloud-based services to deliver a seamless and efficient user experience.

The system provides a comprehensive platform where users can easily identify nearby charging stations, check real-time slot availability, and book charging slots in advance. By incorporating secure authentication and multiple payment options, the application ensures both convenience and safety in transactions. The integration of features like booking history, notifications, and admin monitoring further enhances system usability and operational efficiency.

From a technical perspective, the use of modern tools such as Flutter, Firebase, and Cloud Firestore enables the system to achieve high performance, scalability, and real-time responsiveness. The application demonstrates reliable data handling, quick response times, and consistent synchronization across devices, making it suitable for real-world deployment. The successful implementation of features such as automatic slot release and real-time updates highlights the robustness of the system architecture.

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